

Detailed Referee Report and Priority/Literature Audit

On “A C^1 Counterexample to the Entropy Conjecture”

Submitted manuscript dated 25 September 2026, 15 pages

Author of this referee report: ChatGPT-6

8 October 2026

Scope of review. This report assesses the specific uploaded 15-page article, rather than treating the open problem as an interchangeable generic formulation. Page, equation, proposition and lemma numbers below refer to that article. The external-source investigation was conducted against publicly findable material available by 8 October 2026. A web search cannot certify the absence of unpublished manuscripts, private correspondence, inaccessible grant submissions, or unindexed conference talks. The document below distinguishes verified overlap, conceptual precedents, mathematical deductions made in this report, and unresolved originality questions.

A. Original request (reproduced verbatim)

Be a careful postdoc and write an extensive referee report. Make sure to find all mathematical mistakes and writing suggestions. In particular, list all relevant sources (literature, math SNS, grant proposals available, talk titles and abstracts, and others) related to the attached paper. Assume a significant step is copied somewhere and try hard not to miss one of them. If any of that literature did not appear in the reference section, highlight it in the report. Also, evaluate whether this is a meaningful contribution related to current research progress. Note that it is possible that the counterexample does not improve the mathematical understanding of the math equation we are thinking about.

Include this prompt as a part of the referee report, make the author as "ChatGPT-6", and export to TeX file.

B. Executive assessment and editorial recommendation

Provisional recommendation: major revision and independent expert verification before any publication decision. The claimed theorem is precise and, if correct and genuinely novel, mathematically noteworthy: it gives a noninvertible C^1 map on a closed product manifold with $h_{\text{top}}(f) = 0$ and $\rho(f_*) \geq 2$. It thereby refutes the *all- C^1 -self-maps* formulation of Shub’s homological entropy inequality. It *does not* settle the classical finite-regularity *diffeomorphism* problem. The latter distinction is important in the title, abstract, conclusions and publicity.

I independently followed the central timing/phase/entropy chain and did *not* locate a fatal mathematical inconsistency in the submitted proof. This is not a formal verification and must not be represented as a guarantee of correctness. Two readily obtainable mathematical refinements are omitted: (i) the argument can be strengthened to determine the *entire* homological action and prove $\rho(f_*) = 2$, not merely $\rho(f_*) \geq 2$; (ii) the choice of one exponential amplification constant L may permit an analogous counterexample of any *prescribed finite integer regularity* C^r by adjusting L , provided the higher-order iterate estimates are explicitly supplied. These claims should be included only with rigorous proofs.

Priority/provenance fact: a public `openai/math` repository contains a manuscript with the *same title, manuscript date, principal assertion and matching proof text*, including the

prediction-series and entropy-localization steps; its public source tree also contains the corresponding `entropy.tex`. Its entry in the OpenAI mathematics catalogue is item 151. See § G. This is direct and important textual overlap with a public version of *apparently the same paper*; it is *not* evidence that the manuscript copied a different mathematician’s result, and it does not establish the first-availability date, independence, authorship or priority. No separately authored, older source reproducing the particular finite-register/infinite-prediction construction was identified in the sources inspected. Accordingly, *no plagiarism allegation is warranted on the available evidence*.

C. Mathematical reconstruction and critical checks

C.1. What the argument actually proves

The map is defined on $M = (\mathbb{R}/100\mathbb{Z}) \times \Sigma \times \Sigma^q$, $\Sigma = \mathbb{C} \cup \{\infty\}$, by

$$\begin{aligned} t' &= t + h(t, z) \pmod{100}, \\ z' &= z^2 + 20 \sum_{\ell=1}^q \bar{\beta}_\ell(t) S(v_\ell), \\ v'_\ell &= a(t) S(v_\ell) + U_\ell(t, z), \end{aligned} \tag{1}$$

where S is a radial cutoff, $a = L$ between preparation and readout, and U_ℓ is the infinite prediction series (9). The clock can approach the barrier $t = 50$ only while z contracts to zero; a clock making full periods forces the squaring phase to be matched by one of finitely many amplified registers at a late readout. The authors deduce pointwise attraction to the invariant union $Z_0 \cup Z_\infty$, where the restricted dynamics has zero entropy. Independently, projection to the main sphere intertwines f_* with degree-two squaring on H_2 . This separation of dynamical attraction from global homology is the essential mechanism.

C.2. A systematic proof audit

- M1. Clock feasibility (p. 3, (2)–(3)): checked.** The displayed $b(t) = \eta(t - 50)(t - 50)^2/4 + 1 - \eta(t - 50)$ indeed has its only zero at 50 modulo 100, is 1 outside $(48, 52)$ and satisfies $t + b(t) \leq 50$ for $0 \leq t \leq 50$. Since $h = b + (1 - b)e$, $0 \leq h \leq 1$ and $h = 1$ in the preparation/readout windows. The noncrossing conclusion used in Lemma 3.3 is legitimate.
- M2. Readout plateaux (p. 3, (4)): correct but quantitative omission.** A finite family of supports of length < 1 can cover $[76, 77]$ by their *unit-valued plateaux*. However $q = 1$ is impossible: a single closed unit plateau containing the entire interval $[76, 77]$ would require a supporting interval of length at least 1. Therefore the stated choices force $q \geq 2$ and $\dim M = 2q + 3 \geq 7$ (rather than merely the weaker ≥ 5 on p. 2). State this explicitly and perhaps specify a concrete two- or three-channel covering.
- M3. Radial cutoff p (pp. 4, 13): checked with a needed emphasis.** The requirements $p(r) = r$ for $r \leq 1$, nondecreasing on $[0, 3L]$, bounded by 3, and eventually zero are compatible only because the subsequent decay occurs *beyond* $3L$. The proposed construction does place its final cutoff beyond $3L$. Highlight that this threshold is essential to Lemma 4.2, not an irrelevant technicality.
- M4. Well-definedness at the sphere poles (pp. 4–7, (7), (15)): checked.** S is identity near 0 and zero near ∞ , while U_ℓ vanishes near both poles. For the z -

component, the inverse coordinate is $w' = w^2/(1 + C_0 w^2)$. Boundedness of C_0 gives a common chart neighborhood on which this expression is smooth, so the map extends over $z = \infty$.

- M5. The regularity estimate (p. 5, (11)–(14)): checked at first order.** With $\|DG\| \leq B$ and the angular frequency 2^{m+1} , derivatives of the m th summand are bounded by $CL^{1-m}(1 + B^m + 2^{m+1})$. Thus $L > \max\{B, 2\}$ suffices for uniform convergence of functions and first derivatives. It is important that the B^m and 2^m costs occur in *different* product-rule contributions rather than as an unjustified product $B^m 2^m$. The argument is valid as written for C^1 .
- M6. Periodic seam of U_ℓ (p. 6): checked.** The factor α kills the series near 0 and 100, so nonperiodic future-output tests $\beta_\ell(T_m)$ do not obstruct the 100-periodic C^1 extension. This is a subtle but necessary distinction from the periodized readout coefficients β_ℓ .
- M7. Bounded clock (Lemma 3.2): checked.** Monotonicity gives convergence of a bounded lifted clock. Because $b(t_n) \rightarrow 0$, its limit is 50 modulo 100. The condition $h(t_n, z_n) \rightarrow 0$ excludes $|z_n| \geq 3/4$ eventually, and hence exact squaring forces $z_n \rightarrow 0$. This does not require any convergence of the registers, as the set Z_0 contains *all* register values.
- M8. Reset before preparation (Lemma 3.3): checked.** A clock increment at most one ensures a visit to $[2, 3)$ and hence an update with $a = U_\ell = 0$. The preparation set has at most five lattice-spaced indices, contains an $\alpha = 1$ plateau hit, and cannot have $|z_i| \leq 1/2$ if the clock completes its passage. Use of $\leq 1/2$ in the trap argument appropriately rules out the endpoint.
- M9. Exact prediction of the readout (Lemma 3.4): checked, and central.** Prior to clock 70, the actual (t, z) pair evolves by G because no main-coordinate readout occurs. Over clock readings $[70, 82]$, both actual and predicted clocks advance by exactly one *regardless of their z coordinates*. A fixed readout window has width < 1 , so each register is read at most once. The unperiodized β_ℓ rules out spurious predicted hits in later periods. This should be illustrated in a figure with index conventions.
- M10. Register linearity before a readout (Lemma 3.5): checked.** For a channel read at j , the candidate sum $w_u = L^{u-j} \sum_{i \in I, i < u} c_i$ has size at most $|I|/10 \leq 1/2$, for $i_0 \leq u \leq j$. Since $S(v) = v$ on the unit disk, induction gives the exact readout value without assuming the desired phase alignment. The proof correctly does *not* assume that channels already read remain unsaturated.
- M11. Alignment and radial growth (Lemma 3.6): checked.** Phase θ at i_0 becomes $\theta^{2^{j-i_0+1}}$ in the square at readout j . The U_ℓ exponent 2^{m+1} with $m = j - i$ matches it. The factor $\beta_\ell(s_j)$ arises at *both* preparation and readout, yielding a nonnegative square $\beta_\ell(s_j)^2$. At a plateau hit with $\alpha = \beta = \chi = 1$ this adds 2 to the radius and forces an iterate above 1.
- M12. Global escape (Proposition 3.1): checked.** Once an iterate with modulus > 1 has been produced, every later readout preserves $|z'| \geq |z|^2$ and intervening iterates square exactly. Thus $|z_{N+r}| \geq |z_N|^{2^r} \rightarrow \infty$. Eventually $\chi = 0$, and the next reset kills the register memory. This also treats potentially huge old register values because they cannot act outside their assigned readout window.
- M13. Entropy localization (Lemma 4.1): plausible proof; independent argument available.** The finite-open-cover concatenation shows the uniform-in- x inequality (27)

for the number of exceptional m -blocks, without requiring a uniform last exceptional block. The subsequent encoding yields (28) and zero entropy as $\tau \downarrow 0$ and $m \rightarrow \infty$. A shorter independent proof uses the variational principle: for any invariant probability measure μ , dominated convergence and invariance imply $\int d(x, Z) d\mu(x) = \int d(F^n x, Z) d\mu(x) \rightarrow 0$, so $\mu(Z) = 1$, whence every measure-theoretic entropy is at most $h_{\text{top}}(F|_Z) = 0$. Include this as a cross-check and cite a standard text.

M14. Register entropy (Lemma 4.2): checked. Since R and its iterates are nondecreasing, the N -component radius itineraries form a chain with at most $1 + N(H - 1)$ members, while the angle is preserved. The initial step on the full sphere is correctly accounted for separately. This yields polynomial rather than exponential word complexity.

M15. Homological eigenvalue (Proposition 5.1): checked but can be sharpened. The homotopy $z^2 + sC(t, v)$ is continuous at ∞ by the uniform inverse-coordinate formula. The projection on H_2 is onto, so $P(A - 2I) = 0$ with $P \neq 0$ implies $2 \in \text{spec}(A)$. The proof neither assumes nor proves a dynamical semiconjugacy, which is the correct distinction.

C.3. A strict strengthening: the exact homological spectral radius

The paper proves only $\rho(f_*) \geq 2$, but the entire map f is homotopic to

$$f_0(t, z, v_1, \dots, v_q) = (t, z^2, 0, \dots, 0).$$

Indeed let $C(t, v) = 20 \sum_{\ell} \bar{\beta}_{\ell}(t) S(v_{\ell})$, and define, for $0 \leq s \leq 1$,

$$H_s(t, z, v) = (t + (1 - s)h(t, z), z^2 + (1 - s)C(t, v), (1 - s)[a(t)S(v_{\ell}) + U_{\ell}(t, z)]_{\ell=1}^q). \quad (2)$$

Here the first coordinate is understood modulo 100, and the second at $z = \infty$ by the same uniform inverse-coordinate extension used in (15). The output registers lie in $\mathbb{C}$, so scalar interpolation is continuous; $H_0 = f$ and $H_1 = f_0$. The map f_0 factors through $\mathbb{S}^1 \times \mathbb{S}^2$. By the Künneth decomposition, on real homology the only nonzero eigenvalues are 1 on H_0 , 1 on H_1 , 2 on the distinguished H_2 and 2 on the H_3 generated by circle $\times$ main sphere. All classes using at least one register-sphere factor are killed. Consequently

$$\boxed{\rho(f_*) = 2, \quad h_{\text{top}}(f) = 0, \quad \log \rho(f_*) - h_{\text{top}}(f) = \log 2.}$$

This also shows that the homological eigenvalue 2 occurs in H_3 as well as H_2 , although the chosen H_2 witness remains valid. Equation (2) is a proposed proof addition, not a claim about a computation already present in the submitted paper. It should be checked against the authors' desired orientation and index conventions and incorporated in Section 5.

C.4. Possible extension to arbitrary prescribed finite regularity

The argument appears to be a C^r template for each finite integer r , not an intrinsically C^1 phenomenon. Fix $r < \infty$. Since G is smooth on the compact quotient, repeated chain-rule/Faà di Bruno estimates imply constants A_r, C_r with

$$\max_{0 \leq j \leq r} \|D^j G^m\| \leq C_r A_r^m, \quad \|D^r [\chi(z)(z/|z|)^{2^{m+1}}]\| \leq C'_r (1 + 2^{rm})$$

locally in a finite coordinate atlas. Applied to $\beta_{\ell} \circ T_m$ and multiplied by L^{1-m} , these suggest convergence of all derivative series through order r whenever L exceeds an appropriate r -dependent exponential growth base. The rest of the orbit argument uses only $L > 2$ and the

stated linear/saturation bounds. *Request:* either prove rigorously that for every prescribed finite r a sufficiently large $L = L(r)$ gives a C^r example, or explain a genuine obstruction. Do not infer one fixed C^∞ example: L is fixed for a given map, and exponential decay L^{-m} cannot automatically beat the angular frequency 2^{rm} simultaneously for all r . A verified finite- r extension would significantly improve the precision and importance of the headline theorem.

C.5. Other points requiring clarification, not identified as fatal errors

- (a) State whether q can be minimized, and exhibit explicit plateau intervals; by (4), $q \geq 2$.
- (b) Distinguish C^0 , Lipschitz, C^1 , finite C^r , and C^∞ in every discussion of the historical conjecture.
- (c) Make the step count “signal at i enters v_{i+1} , then has $j - i - 1$ gains before readout at j ” conspicuous, as an off-by-one error here would invalidate the result.
- (d) Name the *unperiodized* future-hit coefficients versus the *periodized* actual readout coefficients distinctly.
- (e) In the abstract, report the precise homological gap, not only the strict inequality, after adding the preceding homotopy.
- (f) Explain that Z_0 is a whole register-space fibre, not a fixed point; only its (t, z) coordinates are fixed.
- (g) Separate pointwise attraction from uniform attraction; the former suffices by variational principle, the latter is not claimed.
- (h) Consider whether the construction can be performed in smaller dimension or with fewer clock/register features; this is relevant to how broad a negative result has actually been obtained.

D. Significance and limits relative to ongoing research

What would be new: on the evidence examined, the combination of C^1 regularity, a closed manifold, an intermediate-degree homology eigenvalue > 1 , and vanishing entropy is not covered by the familiar Lipschitz/ C^0 counterexamples or by degree obstructions for C^1 maps. The noninvertibility and intermediate homology are used for a reason, not merely to evade terminology: known H_1 lower bounds already hold for continuous maps, and the top-degree bound holds for C^1 maps.

What would not be new or resolved: this is *not* a counterexample among C^1 diffeomorphisms; nor does it refute the valid C^∞ theorem of Yomdin. The traditional motivation concerns the least possible entropy in a *diffeomorphism isotopy class*; the present construction may say rather little about that continuing problem. One should not advertise “Shub’s entropy conjecture is false” without the restrictive phrase “in its general C^1 self-map formulation.” The literature continues to treat finite-regularity diffeomorphism cases, such as conservative and partially hyperbolic classes, as active problems.

Conceptual contribution: the novel-looking technical idea is a clock slowed near a trapping section, paired with an absolutely convergent, infinitely indexed *prediction series* and finitely many gain/reset registers. This arrangement converts possibly unbounded waiting time into a compulsory phase-aligned forcing contribution while retaining first-derivative control. Once the attraction statement is available, the entropy localization and homology facts are comparatively

standard. This is a real construction result but does not establish a more general dynamical principle, classification, sharp dimension threshold, or insight into the diffeomorphism case. Its importance would rise if the authors proved the proposed finite- r strengthening, optimized dimension, or formulated a general obstruction/realization theorem explaining which homotopy classes admit zero-entropy C^r noninvertible representatives.

Recommended decision criteria: correctness after specialist checks of (9), (19), (21), (25), and (27); full disclosure and contextualization of the public preprint/source version; a detailed comparison with earlier C^0 /Lipschitz counterexamples; and an honest statement of the unresolved diffeomorphism problem. Without the last two, the headline risks exaggerating impact even if the theorem is correct.

E. Literature that appears relevant but is absent from the paper's 11-item reference list

The existing references [1]–[11] already include Gromov, Hatcher, Liao–Viana–Yang, two Manning items, Misiurewicz on the variational principle, Misiurewicz–Przytycki on degree, Saghin–Xia, two Shub items, and Yomdin. Those have not been mislabeled “missing.” The following independently traceable items are *not* listed in that bibliography. Priority in this section is thematic; a paper relevant to the background is *not* thereby a source for the new clock construction.

- L1. Not cited in submitted manuscript. C. C. Pugh, “On the entropy conjecture: a report on conversations among R. Bowen, M. Hirsch, A. Manning, C. Pugh, B. Sanderson, M. Shub, and R. Williams,”** in *Dynamical Systems—Warwick 1974*, LNM 468 (1975), 257–261. *High priority:* the primary historical source for Lipschitz counterexamples mentioned indirectly on p. 1. Bibliographic record in Shub’s ICM references: https://shub.ccny.cuny.edu/articles/2006-All_Most_Some_Differentiable_Dynamical_Systems.pdf.
- L2. Not cited in submitted manuscript. A. Katok, “A conjecture about entropy,”** 1986, cited and discussed in subsequent surveys of the C^0 counterexample on the two-sphere. *High historical relevance, bibliographic details to verify from the original text:* see <https://arxiv.org/abs/1209.3936> and <https://www.mdpi.com/2227-7390/9/4/296>. Do not treat the secondary claim about Katok’s construction as proof of any overlap with (9).
- L3. Not cited in submitted manuscript. M. Misiurewicz and F. Przytycki, “Entropy conjecture for tori,”** *Bull. Acad. Polon. Sci.* **25** (1977), 575–578. *High:* shows continuous torus maps satisfy a positive entropy bound, helping explain the manifold choice. Source: <https://www.impan.pl/~feliksp/publ.pdf>.
- L4. Not cited in submitted manuscript. W. Marzantowicz and F. Przytycki, “Entropy conjecture for continuous maps of nilmanifolds,”** *Israel J. Math.* **165** (2008), 349–379. *High:* a whole class of spaces on which arbitrary continuous maps obey the inequality. <https://doi.org/10.1007/s11856-008-1015-0>.
- L5. Not cited in submitted manuscript. W. Marzantowicz and F. Przytycki, “Estimates of the topological entropy from below for continuous self-maps on some compact manifolds,”** *Discrete Contin. Dyn. Syst.* **21** (2008), 501–512. *High:* extension to infra-nilmanifolds and virtually nilpotent fundamental groups; arXiv:0705.1610. <https://doi.org/10.3934/dcds.2008.21.501>.

- L6. Not cited in submitted manuscript. D. Fried, “Entropy and twisted co-homology,”** *Topology* **25** (1986), 455–470. *High conceptual relevance:* unified and strengthened homological conjectures; a more precise comparison of which algebraic invariants survive the example is warranted. [https://doi.org/10.1016/0040-9383\(86\)90024-8](https://doi.org/10.1016/0040-9383(86)90024-8).
- L7. Not cited in submitted manuscript. D. Fried and M. Shub, “Entropy, linearity and chain-recurrence,”** *Publ. Math. IHÉS* **50** (1979), 203–214. *Medium–high:* global algebraic complexity versus recurrent-set structure is central to the paper’s interpretation. https://www.numdam.org/item/PMIHES_1979__50__203_0/.
- L8. Not cited in submitted manuscript. M. Shub and R. F. Williams, “Entropy and stability,”** *Topology* **14** (1975), 329–338. *High for context:* positive results for Axiom A/no-cycles diffeomorphisms. <https://profshub.github.io/publications.html>.
- L9. Not cited in submitted manuscript. D. Ruelle and D. Sullivan, “Currents, flows and diffeomorphisms,”** *Topology* **14** (1975), 319–327. *Medium–high:* dynamical growth and homological currents in hyperbolic settings. [https://doi.org/10.1016/0040-9383\(75\)90016-6](https://doi.org/10.1016/0040-9383(75)90016-6).
- L10. Not cited in submitted manuscript. R. Sacksteder and M. Shub, “Entropy of a differentiable map,”** *Adv. Math.* **28** (1978), 181–185; and **“Entropy on sphere bundles,”** *Adv. Math.* **28** (1978), 174–177. *Medium:* prior entropy/homology studies, the latter especially because the example uses products of spheres. <https://profshub.github.io/publications.html>.
- L11. Not cited in submitted manuscript. S. Newhouse, “Entropy and volume,”** *Ergodic Theory Dynam. Systems* **8**, special issue (1988), 283–299. *High for finite regularity/volume context:* complement to Yomdin’s C^∞ lower bound, but not a construction of the manuscript’s counterexample. https://www.cambridge.org/core/services/aop-cambridge-core/content/view/6A86AB918E9AE957D9D1CC5F17F1CD02/S0143385700009469a.pdf/entropy_and_volume.pdf.
- L12. Not cited in submitted manuscript. R. Bowen, “Entropy and the fundamental group,”** in *The Structure of Attractors in Dynamical Systems*, LNM 668 (1978), 21–29. *Medium:* first-homotopy-level lower bounds for continuous maps, complementing Manning. Cited in the references of https://ddd.uab.cat/pub/prepub/2008/hd1_2072_12815/Pr808.pdf.
- L13. Not cited in submitted manuscript. P. Liu and K. Lu, “A note on partially hyperbolic attractors: Entropy conjecture and SRB measures,”** *Discrete Contin. Dyn. Syst.* **35** (2015), 341–352. *Medium:* another positive restricted class, with a dynamically natural hypothesis absent from the example. <https://doi.org/10.3934/dcds.2015.35.341>.
- L14. Not cited in submitted manuscript. L. Luo, “The Topological Entropy Conjecture,”** *Mathematics* **9** (2021), article 296, with an earlier 2012 preprint arXiv:1209.3936. *Medium, qualifying caution:* this paper changes the entropy invariant to a *fiber entropy*, and hence does not prove the inequality disputed here. Useful for historical context only; distinguish invariant definitions. <https://doi.org/10.3390/math9040296>.
- L15. Not cited in submitted manuscript. M. Shub and D. Sullivan, “Homology theory and dynamical systems,”** *Topology* **14** (1975), 109–132. *Medium:* homolog-

ical dynamics and fixed-point constraints. <https://profshub.github.io/publications.html>.

- L16. Not cited in submitted manuscript.** M. Misiurewicz and F. Przytycki, “Entropy conjecture for tori” and the nilmanifold papers above should be used to explain why the present example is not simply a toral construction. This is a synthesis instruction, not an additional distinct citation.

Literature priority: [L1], [L3]–[L6] and [L11] merit explicit citation or discussion in a revised introduction. [L7]–[L10], [L12]–[L15] may be grouped in a concise paragraph or notes depending on journal space. A bibliography should not be inflated merely for exhaustiveness; missing direct precursors matter more than tangential papers.

F. Math SNS, talk abstracts and public research projects

These records are especially helpful for assessing how the conjecture was understood *before* this manuscript; none of the sources inspected here displays the exact preparation/readout series independently of the public version of the submitted article.

- W1. Not cited in submitted manuscript.** Rufus Bowen’s Notebook, Problem 12, “Shub’s entropy conjecture”; public comments attributed to Ledrappier (2016) and Shub (2017). Records positive Axiom A/no-cycles and C^∞ results and the continuing diffeomorphism-oriented framing. <https://bowen.pims.math.ca/problem/problem-12/>. *Research-community problem list, not an original proof of the present construction.*
- W2. Not cited in submitted manuscript.** MathOverflow, “Shub Conjecture and polynomial entropy” (2024). Questions whether homological growth has a subexponential/polynomial-entropy analogue; a comment explicitly asks about f_* nonzero with $h_{\text{top}}(f) = 0$. <https://mathoverflow.net/questions/474753/shub-conjecture-and-polynomial-entropy>. *Conceptual overlap only; the question does not specify an eigenvalue of modulus > 1 nor the present mechanism.*
- W3. Not cited in submitted manuscript.** MathOverflow, “Entropy conjecture for flows” (2010). Discusses the diffeomorphism version and global topological entropy obstructions, with links to Katok/Shub surveys. <https://mathoverflow.net/questions/39789/entropy-conjecture-for-flows>. *Background, low priority.*
- W4. Not cited in submitted manuscript.** Andrea Ulliana, “The Topological Entropy Conjecture,” Mathematics in Conversation / University of Padua seminar, 11 June 2024. Public abstract explicitly says the finite-regularity *diffeomorphism* problem was open, with positive results under restrictions. <https://www.math.unipd.it/news/the-topological-entropy-conjecture/>; also <https://sites.google.com/view/mathematicsinconversation/past-seminars/20232024-seminars>. *Talk title and abstract; not a proof publication.*
- W5. Not cited in submitted manuscript.** Axel Kodat, “An introduction to the entropy conjecture,” CUNY Graduate Center seminar, 8 December 2023. Abstract starts with a smooth endomorphism of a closed manifold, recalls Yomdin and describes finite regularity as widely open. https://tamara.ccny.cuny.edu/Seminar_F2023.html.
- W6. Not cited in submitted manuscript.** Axel Kodat, “Entropy and local volume

growth for C^2 conservative diffeomorphisms,” CUNY Graduate Center seminar, fall 2025. Public abstract announces further partial results for conservative C^2 diffeomorphisms, highlighting a currently active regime not challenged by this map. https://tamara.ccny.cuny.edu/seminar_F2025.html. No manuscript/preprint containing the announced proof was verified here.

W7. Not cited in submitted manuscript. W. Marzantowicz and F. Przytycki, “Estimates of the topological entropy from below for continuous self-maps on some compact manifolds,” slides hosted at Jagiellonian University (23 May 2011). <https://dynsys.im.uj.edu.pl/old/briefs/2011-01-21.pdf>. These slides reflect the established algebraic classes and do not disclose the new delay-register trick.

W8. Not cited in submitted manuscript. European Commission CORDIS, FP7 project 234559, “Smooth Ergodic Theory and Partially Hyperbolic Diffeomorphisms.” A *public project description/proposal-style abstract* discusses homology classes of stable/unstable foliations, disk-volume growth and entropy, and explicitly names Shub–Williams, Ruelle–Sullivan, and Saghin–Xia. <https://cordis.europa.eu/project/id/234559>. This is a substantiated, directly relevant publicly accessible grant record; it does not prove the new example or suggest appropriation from the project.

W9. Not cited in submitted manuscript. National Natural Science Foundation of China, grant 11801428, acknowledged in L. Luo (2021). A public *grant acknowledgement*, not an available project proposal or a demonstrated connection to the prediction mechanism. <https://www.mdpi.com/2227-7390/9/4/296>. Do not falsely describe the grant acknowledgement as a full proposal.

Search limitation on grants: no publicly accessible funding proposal reproducing the register construction or the proposed main theorem was established. Generic grants on entropy and dynamical systems were excluded, because matching a keyword alone is not evidence of intellectual dependence.

G. Specific overlap / provenance investigation

G1: very high-confidence correspondence to a public copy of this manuscript. The OpenAI mathematical repository contains an `article.pdf` under the path

<https://github.com/openai/math/blob/main/preprints/A-C1-Counterexample-to-the-Entropy-Conjecture-September-25-2026/article.pdf>

and corresponding editable source at

<https://github.com/openai/math/blob/main/preprints/A-C1-Counterexample-to-the-Entropy-Conjecture-September-25-2026/build/source/entropy.tex>.

The repository catalogue <https://github.com/openai/math/blob/main/overview.tex> lists this as item 151. The title, abstract, statement, clock/register formulas, proof of the localization lemma and other visible textual passages match the supplied article. The third-party catalogue <https://kisonecat.github.io/openai-math/dynamical-systems-and-ergodic-theory.html> independently indexes the same public manuscript, but *not* as an independent mathematical proof. This is a significant exact-source correspondence and should

be disclosed as the public source/earlier version if the manuscript is resubmitted anonymously. It does not show that a substantive step was copied *from somebody else's earlier publication*. The public repository page does not by itself establish when the content first became accessible; a documented revision history or archival identifier would be needed to assign priority dates.

G2: possible thematic origins requiring citation, but no demonstrated derivation.

The *homology-versus-entropy* idea is as old as Shub/Pugh and Fried; the inability of smooth degree to produce zero entropy is described by Misiurewicz–Przytycki; the localization argument follows readily from the variational principle; and phase amplification by a linear register is elementary once the prediction trick is specified. These are mathematically transparent relationships, *not evidence that formulas (9), (19), (21) or (25) appear in [L1]–[L15] or [W1]–[W9]*. The source that would matter most for priority is an independently dated presentation of the *infinite future-hit prediction sum* with coefficients L^{1-m} , frequencies 2^{m+1} , and the “once per passage” readout identity. None was verified in this audit.

G3: suggested priority procedure. Request an explicit public-version statement from the submitting author; check repository commit history and archived snapshots; query subject experts for independent unpublished versions; and ask the author to identify which lemmas are new and which were borrowed from folklore. Any allegation of misconduct must be based on traceable independent sources and chronology, not on the reviewer’s instruction to *assume* copying. Open access to the apparent original version is a transparent provenance fact, not a verdict of plagiarism.

H. Detailed writing and presentation improvements

- E1. Title/abstract:** prefer “A noninvertible finite-regularity counterexample to the general homological entropy inequality” or explicitly include *self-map*. The current title is not false but invites confusion with the famous diffeomorphism version.
- E2. First paragraph:** separate “Shub’s original diffeomorphism conjecture,” “its later endomorphism formulation,” and “the weaker regularity counterexamples.” Cite Pugh’s original report, not only Shub’s 2007 retrospective.
- E3. State the strongest exact result:** if the homotopy in § C.3 is accepted, write $\rho(f_*) = 2$, give the exact gap $\log 2$, and mention the additional H_3 eigenvalue.
- E4. Dimension:** explain $q \geq 2$ and provide a simple concrete choice of supports and plateaux; report the resulting lower bound $\dim M \geq 7$ for this method.
- E5. A readable schematic:** show the two time axes “clock reading” and “iteration index” side-by-side; long delays near 50 do *not* mean large spatial motion. State that $m = j - i$ is an iterate count.
- E6. Equation (9):** describe $\Phi_{2^{m+1}}(z) = \chi(z)(z/|z|)^{2^{m+1}}$ separately to avoid confusion between 2^{m+1} and $2m + 1$ when the paper is converted to text.
- E7. Lemma 3.4:** elevate exact identification (20) of predicted and actual clocks to an explicit subclaim with a short proof; this is the lynchpin connecting prediction to actual forcing.
- E8. Lemma 3.5:** keep the linearity verification *before* phase induction. Emphasize the estimate $|w_u| \leq 1/2 < 1$ and the role of resetting the registers.
- E9. Lemma 3.6:** explain why the coefficient is $2 \sum_{\ell} \beta_{\ell}(s_j)^2 \sum_{i \in I} \alpha(s_i) \chi(z_i)$, which shows nonnegativity directly.

- E10. Entropy Section 4:** state the one-paragraph invariant-measure proof before or after the direct separated-set lemma; explain why pointwise eventual nearness suffices without an attraction rate.
- E11. Definitions:** choose one notation for Σ , $\bar{\beta}_\ell$, Z_0 , Z_∞ , and ρ ; write all powers 2^{m+1} unmistakably; avoid calling d_m merely an “orbit pseudometric” since including time zero makes it a metric.
- E12. Discuss smoothness threshold:** either prove the finite- r extension suggested above or note explicitly why the chosen L bounds only finitely many derivatives and cannot support the C^∞ theorem’s hypotheses.
- E13. Scope paragraph in conclusion:** add one paragraph on why noninvertibility, failure of surjectivity if applicable, and the topology of products of spheres limit relevance to the diffeomorphism conjecture. Do not infer nonsurjectivity without proving it.
- E14. Bibliography:** add the priority references [L1], [L3]–[L6], [L11]; check author-name diacritics and consistent capitalization of C^r , C^∞ , *Astérisque*, and the journal titles. Include a version identifier for the repository-hosted manuscript.
- E15. Literature map:** a short table distinguishing *continuous maps* / *Lipschitz maps* / C^1 *endomorphisms* / C^1 *diffeomorphisms* / C^∞ *maps*, with known positive and negative results, would be more informative than the current compressed historical paragraph.

I. Final verdict

I find a serious-looking but internally coherent constructive argument, not a documented fatal disproof. The claims that merit independent checking are the infinite derivative bound for the prediction series, the exact synchronization through successive readouts, the unsaturated-register induction, and the move from pointwise attraction to zero entropy. The last point has an especially simple independent verification by invariant measures. I recommend **major revision**, with the two homological and regularity-strengthening questions treated as substantive mathematical requests, the missing primary historical sources corrected, and the overlap with the public OpenAI repository transparently documented. Acceptance as a resolution of the *diffeomorphism* conjecture would be mathematically unjustified. A correct new example for noninvertible C^1 self-maps remains a meaningful negative result, but its direct explanatory impact on the central diffeomorphism question is limited.

Verification note. Bibliographic URLs and public records are included for auditability, but links may change. The mathematical refinements in §C.3–C.4 are the referee’s deductions and proposals, not claims established by the external webpages. Search completeness and the originality of an unpublished construction cannot be certified from web-indexed material alone.